CONFIGURING BASIC NETWORK HARDWARE THEORY

Practical Assignment

Topics covered

- Router administration and login page
- IP address, subnet mask, and default gateway
- DHCP and static IP configuration
- Testing network connectivity
- DHCP range and IP address conflicts
- Troubleshooting and safety precautions

Note: Fill in the actual values from your own home Wi-Fi network. Example IP addresses in this assignment are illustrative only.

Introduction

A home network allows devices such as laptops, phones, smart TVs, and printers to communicate and access the internet. A Wi-Fi router usually connects the home network to the internet and may also provide wireless access, routing, a firewall, and DHCP address assignment. Understanding basic network settings helps users configure devices safely and troubleshoot common connection problems.

Objectives

After completing this practical, a student should be able to:

- Open the router's local administration page.
- Identify the device IP address, subnet mask, and default gateway.
- Explain the difference between automatic DHCP and a static IP address.
- Test a static IP configuration and restore DHCP afterwards.
- Find the DHCP address pool in the router settings.
- Explain what happens when two devices use the same IP address.

1. Open the Router Admin Login Page

A router admin page is a local web page used to manage network settings. It may let an authorised administrator view Wi-Fi settings, change the Wi-Fi password, check connected devices, update firmware, and manage DHCP. The exact page and menu names depend on the router brand and model.

Procedure

1. Connect your laptop or phone to your home Wi-Fi network.
2. Open a web browser.
3. Enter your router's local address. Common examples are **192.168.0.1** and **192.168.1.1**, but your router may use a different address.
4. If the page does not open, check the Default Gateway in your device's network settings and enter that address into the browser.
5. When the router login page appears, take a screenshot for your practical record.

Finding the router address on Windows

Open Command Prompt and type **ipconfig**. Find the Wi-Fi adapter and read the **Default Gateway** value. Enter that IP address into the browser's address bar. Do not search for the address in a search engine; type it directly into the address bar.

Screenshot record

Paste or attach a screenshot of your own router login page here. Do not include or share your router password, recovery code, or other private information.

2. Find Your Device’s Network Settings

Each device connected to a local network usually has an IP address. The subnet mask helps identify the local network, and the default gateway is normally the router that forwards traffic to other networks, including the internet.

Setting	Meaning	Value from your device
IPv4 address	The local address assigned to your laptop or phone.	
Subnet mask	Shows which part of the IPv4 address identifies the network.	
Default gateway	Usually the router’s local IP address.	
DNS server (optional)	Helps translate website names into IP addresses.	

Windows method

1. Press the Windows key, type **cmd**, and open Command Prompt.
2. Type **ipconfig** and press Enter.
3. Under the Wi-Fi adapter, record IPv4 Address, Subnet Mask, and Default Gateway.
4. If several adapters appear, use the one currently connected to Wi-Fi.

Mobile method

Open Settings, choose Wi-Fi, tap the connected network, and look for network details or IP settings. The exact labels depend on the phone model and operating system.

Example only

A device might have IP address **192.168.1.25**, subnet mask **255.255.255.0**, and default gateway **192.168.1.1**. These are examples, not values to copy blindly. Record the actual settings from your own network.

Why these values matter

The IP address identifies a device on the local network. The subnet mask helps the device decide whether another address is on the same local network. The default gateway forwards traffic to destinations outside the local network. Incorrect values can prevent access to the router or internet.

3. Understand DHCP and Static IP Addresses

DHCP stands for Dynamic Host Configuration Protocol. A DHCP server, commonly built into a home router, automatically gives devices network settings such as an IP address, subnet mask, default gateway, and sometimes DNS servers. This makes connecting new devices easier.

Automatic (DHCP): The device requests its network settings automatically. This is the recommended setting for most home users.

Static IP: The user enters the IP settings manually. A static address can be useful for certain devices, but it must be selected carefully to avoid conflicts.

4. Test a Static IP Address

Before changing anything, write down or photograph your current network settings. Only change settings on a device and network you are authorised to manage. For this practical, use an address that belongs to the same subnet and is not already used by another device.

Example

If the router address is **192.168.1.1** and the subnet mask is **255.255.255.0**, an example static address could be **192.168.1.50**. This is only suitable if the address is unused and does not conflict with the DHCP pool or any reserved address. Use your actual router and subnet information, not this example by default.

Windows configuration

1. Open Settings.
2. Select Network & Internet, then Wi-Fi.
3. Open the properties of the connected Wi-Fi network.
4. Find IP assignment and choose Edit.
5. Select Manual and enable IPv4.
6. Enter a suitable IP address, subnet mask or prefix length, default gateway, and DNS settings.
7. Save the settings and test whether you can open a website.

The wording and location of these menus can differ by Windows version. If you are unsure which address is safe to use, do not guess; leave DHCP enabled or ask the network administrator.

Testing and restoring DHCP

After saving the static configuration, check whether the router is reachable and whether websites load. If the connection fails, restore **Automatic (DHCP)** immediately. After the test, return IP assignment to Automatic (DHCP) even if the static address worked, as required by the assignment.

5. Find the Router’s DHCP Range

The DHCP range, also called the DHCP pool, is the group of IP addresses the router can automatically assign to devices. The range is usually shown in the router administration page under a menu such as LAN, Local Network, or DHCP Server. The exact menu depends on the router.

Procedure

1. Open the router’s admin page and sign in if authorised.
2. Look for LAN, Local Network, or DHCP Server settings.
3. Find the DHCP start address and end address.
4. Record the values below. Do not change them unless your practical specifically requires it and you understand the effect.

DHCP information	Value from your router
DHCP start address	
DHCP end address	
Router / gateway address	

Example DHCP range

A router might assign addresses from **192.168.1.100** through **192.168.1.200**. This is only an example. Your router may use a different range and subnet.

6. What Happens if Two Devices Have the Same IP Address?

If two devices use the same IP address on the same local network, an IP address conflict can occur. Network traffic may reach the wrong device or fail, and one or both devices may lose reliable network access. The devices may show connection warnings or have trouble reaching the router and internet.

A DHCP server normally tries to avoid assigning the same address to two devices. However, a conflict can happen if someone manually configures a static IP that is already in use or overlaps with the DHCP pool. A static address should be selected carefully and coordinated with the network administrator.

7. Troubleshooting Common Problems

Problem	Possible reason / safe action
Router login page will not open	Check that you are connected to the correct Wi-Fi and use the Default Gateway shown by your devi
No internet after static IP	Restore Automatic (DHCP). The address, gateway, subnet, or DNS setting may be incorrect.
IP conflict warning	Return to DHCP or ask the network administrator to select an unused address.

Problem	Possible reason / safe action
Cannot find DHCP range	Look under LAN, Local Network, or DHCP Server; menu names differ by router.
Forgot router admin password	Check the router label or official manual, or ask the owner/administrator. Avoid resetting a router wit

8. Practical Observation Sheet

Fill in the results after performing the practical on your own network. Do not copy example values as if they were your actual results.

Observation	Result
Router login page opened?	Yes / No
Device IPv4 address	
Subnet mask	
Default gateway	
DHCP start address	
DHCP end address	
Static IP test result	Internet worked / did not work
Restored Automatic (DHCP)?	Yes / No

9. Review Questions and Answers

Q1. What is a router?
A router connects different networks and forwards traffic between them. A home router commonly connects the local home network to the internet.

Q2. What is an IP address?
An IP address is a logical network address used to identify a device or interface so network traffic can be delivered to it.

Q3. What is a subnet mask?
An IPv4 subnet mask identifies which part of an address belongs to the network and which part identifies a host within that network.

Q4. What is a default gateway?
It is the device, usually the router, that forwards traffic from the local network to other networks.

Q5. What is DHCP?
DHCP automatically provides devices with IP configuration, reducing the need to enter settings manually.

Q6. Why must DHCP be restored after the test?
Returning to Automatic (DHCP) avoids leaving the device with incorrect or conflicting manual settings and allows the router to configure it normally.

Q7. What is an IP conflict?
It occurs when two devices on the same network use the same IP address, which can interrupt

communication.

Conclusion

In this practical, the steps for opening a router login page, checking IP configuration, testing a static IP, and finding the DHCP range were studied. These skills help explain how devices communicate on a home network and how to identify basic network problems. The device should always be returned to Automatic (DHCP) after the static-IP test.

Safety reminders

- Only access a router you own or are authorised to manage.
- Do not share router passwords or screenshots containing private details.
- Do not change settings you do not understand.
- Restore Automatic (DHCP) after the static IP test.